Correcting Forward Model Mismatch in Coded Aperture Snapshot Spectral Imaging via Two-Stage Differentiable Calibration

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Abstract. Coded aperture snapshot spectral imaging (CASSI) captures a 3D hyperspectral cube from a single 2D measurement using a coded mask and spectral dispersion. Deep learning reconstructors such as MST achieve state-of-the-art quality (>34 dB) but assume perfect knowledge of the forward operator. In practice, sub-pixel mask misalignment (Δx, Δy) and rotation (θ) between the coded aperture and detector are unavoidable, yet even 1.5-pixel shift degrades MST-L reconstruction by 16.40 dB (from 34.81 to 18.41 dB). We propose a two-stage differentiable calibration pipeline: (1) a coarse hierarchical grid search over 567 candidate parameter triplets scored by GPU-accelerated GAP-TV, followed by (2) joint gradient refinement through an unrolled differentiable forward operator using a Straight-Through Estimator (STE) for integer dispersion offsets. The pipeline is self-supervised, requiring only the measurement and nominal mask—no ground truth scene. On 10 KAIST benchmark scenes with injected mismatch (Δx=1.5, Δy=1.0, θ=0.3°), our method recovers +3.01 dB for MST-L and +2.99 dB for PnP-HSICNN through self-supervised calibration. We evaluate five reconstruction methods (GAP-TV, MST-S, MST-L, HDNet, PnP-HSICNN) across four scenarios, revealing a mask-sensitivity spectrum: mask-guided transformers suffer catastrophic degradation (−16 dB) but gain most from calibration (+3 dB), while deep prior methods (HDNet) show inherent robustness (−8.8 dB) with smaller calibration benefit. We release all code, results, and a standardized four-scenario evaluation framework.

Keywords: CASSI · Operator mismatch · Differentiable calibration · Straight-Through Estimator · Hyperspectral imaging

1 Introduction

Coded aperture snapshot spectral imaging (CASSI) [11,1] captures a 3D hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times \Lambda}$ from a single 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times W'}$ through the combined action of a binary coded mask and a dispersive element. The forward model is

$$y(i,j) = \sum_{k=1}^{\Lambda} m(i,j) \cdot x(i, j - d_k, k) + n(i,j), \tag{1}$$

where m is the coded aperture, $d_k = k \cdot s$ is the spectral dispersion offset for band k with stride s, and n is measurement noise.

Recent deep learning approaches—particularly Mask-guided Spectral-wise Transformers (MST) [3,4]—achieve remarkable reconstruction quality (>34 dB PSNR on the KAIST benchmark [5]) by jointly processing the measurement and the known mask pattern. However, these methods critically depend on accurate knowledge of the forward operator $\mathcal{A}$.

The mismatch problem. In deployed CASSI systems, the actual mask position inevitably differs from the assumed position due to manufacturing tolerances, assembly errors, and thermal drift. Specifically, three parameters characterize the dominant mask-detector misalignment: horizontal shift Δx, vertical shift Δy, and rotation angle θ. Even modest mismatches ($\Delta x = 1.5\,\text{px}$, $\Delta y = 1.0\,\text{px}$, $\theta = 0.3°$) degrade MST-L reconstruction by 16.40 dB—from 34.81 dB to 18.41 dB—rendering the system effectively unusable. In contrast, deep prior methods like HDNet [6] suffer only 8.79 dB degradation (34.79 to 26.00 dB) due to their learned spectral priors, while classical methods like GAP-TV [13] show only 0.40 dB degradation but at lower peak quality (20.37 dB).

Challenges in CASSI calibration. Correcting mask mismatch through the reconstruction pipeline presents unique challenges:

1. **Integer dispersion**: The spectral dispersion d_k maps to integer pixel offsets, creating a non-differentiable forward operator.
2. **Coupled parameters**: Translation and rotation interact through the mask pattern, making sequential optimization suboptimal.
3. **No ground truth**: In practice, neither the true mismatch parameters nor the ground truth scene are available—calibration must be self-supervised from the measurement alone.

Contributions. We address these challenges with:

1. A **differentiable CASSI forward model** using a Straight-Through Estimator (STE) [2] for integer dispersion offsets, enabling gradient-based calibration.
2. A **two-stage calibration pipeline**: coarse grid search (Stage 0–1) followed by gradient refinement (Stage 2A–2C) for robust, sub-pixel parameter recovery.
3. A **self-supervised objective**: measurement residual minimization requires only the measurement $\mathbf{y}$ and nominal mask $\mathbf{m}$—no ground truth scene or external calibration targets.
4. A **four-scenario evaluation framework** (Ideal, Assumed, Corrected, Oracle) that systematically quantifies mismatch degradation, calibration recovery, and residual gap across reconstruction methods.

2 Related Work

CASSI reconstruction. Classical approaches including GAP-TV [13,8] use alternating projection with total variation regularization. Plug-and-play methods such as PnP-HSICNN [14] combine optimization frameworks (ADMM/GAP) with learned denoisers. Deep learning methods have significantly advanced quality: HDNet [6] uses dual-domain deep unfolding, while MST [3] introduces mask-guided spectral-wise attention achieving 35+ dB on KAIST. All assume perfect forward operator knowledge.

Operator-aware reconstruction. Physics-based learned design [7] optimizes optical elements end-to-end but requires differentiable forward models. Deep unrolling approaches [10,15] embed the forward operator into network layers, making them sensitive to operator errors. HyperReconNet [12] jointly optimizes mask design and reconstruction but does not address post-fabrication calibration.

Self-calibration in computational imaging. Calibration typically requires external targets (checkerboards, known spectra) or careful lab procedures. Self-calibration from measurements alone has been explored for phase retrieval [9] but not for CASSI mismatch correction with deep reconstructors.

Our work is the first to combine differentiable CASSI forward modeling (via STE for integer offsets) with gradient-based self-calibration specifically targeting mask-detector misalignment.

3 Problem Formulation

3.1 CASSI Forward Model

The SD-CASSI (single-disperser) forward model maps a hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H\times W\times \Lambda}$ to a 2D measurement $\mathbf{y} \in \mathbb{R}^{H\times(W+(\Lambda-1)s)}$:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}; \mathbf{m}, \{d_k\}) = \sum_{k=1}^{\Lambda} \text{shift}_{d_k}(\mathbf{m} \odot \mathbf{x}_k) + \mathbf{n}, \tag{2}$$

where $\mathbf{m} \in \{0,1\}^{H\times W}$ is the coded aperture, $d_k = k \cdot s$ is the integer dispersion offset for band k, s is the stride (typically 2), and shift_{d_k} shifts the column index by d_k pixels.

3.2 Mismatch Parameterization

We model mask-detector misalignment as a 3-parameter affine transformation of the mask:

$$\tilde{\mathbf{m}} = \mathcal{W}(\mathbf{m}; \Delta x, \Delta y, \theta), \tag{3}$$

where $\mathcal{W}$ applies bilinear-interpolated translation $(\Delta x, \Delta y)$ and rotation θ about the mask center. The true measurement is generated with the misaligned mask $\tilde{\mathbf{m}}$, while reconstruction assumes the nominal mask $\mathbf{m}$.

3.3 Calibration Objective

Given measurement $\mathbf{y}$ (generated with unknown true mismatch $\boldsymbol{\psi}^* = (\Delta x^*, \Delta y^*, \theta^*)$) and nominal mask $\mathbf{m}$, we seek:

$$\hat{\boldsymbol{\psi}} = \arg\min_{\boldsymbol{\psi}} \left\| \mathbf{y} - \mathcal{A}\big(\mathcal{R}(\mathbf{y}, \mathcal{W}(\mathbf{m}; \boldsymbol{\psi})); \mathcal{W}(\mathbf{m}; \boldsymbol{\psi}), \{d_k\}\big) \right\|^2, \tag{4}$$

where $\mathcal{R}(\mathbf{y}, \tilde{\mathbf{m}})$ is a reconstruction algorithm (GAP-TV in our pipeline) that produces a spectral cube estimate from measurement $\mathbf{y}$ using mask $\tilde{\mathbf{m}}$. This is self-supervised: minimizing the measurement residual requires no ground truth.

4 Method

4.1 Differentiable CASSI Forward Model

The key challenge is that dispersion offsets $d_k = k \cdot s$ are integers, making shift_{d_k} non-differentiable. We address this with a Straight-Through Estimator (STE) [2]:

$$\hat{d}_k = \text{round}(d_k), \quad \frac{\partial \hat{d}_k}{\partial d_k} \equiv 1. \tag{5}$$

In the forward pass, offsets are rounded to integers for exact indexing; in the backward pass, gradients flow through as if rounding were the identity function. This enables gradient-based optimization of parameters that influence the dispersion model.

The differentiable mask warping $\mathcal{W}(\mathbf{m}; \boldsymbol{\psi})$ uses PyTorch's `affine_grid` and `grid_sample` with bilinear interpolation, providing exact gradients for Δx, Δy, and θ. The sign convention matches scipy exactly: $t_x = -2\Delta x/W$, $t_y = -2\Delta y/H$.

4.2 Differentiable GAP-TV Solver

We unroll K iterations of GAP-TV into a differentiable computation graph:

$$\mathbf{r}^{(t)} = \mathbf{y} - \mathcal{A}(\mathbf{x}^{(t)}; \tilde{\mathbf{m}}, \{d_k\}), \tag{6}$$

$$\mathbf{x}^{(t+1)} = \text{TV}_\sigma\big(\mathbf{x}^{(t)} + \mathcal{A}^\dagger(\mathbf{r}^{(t)})\big), \tag{7}$$

where TV_σ denotes Gaussian-weighted TV denoising (replacing the standard TV proximal step for differentiability), and $\mathcal{A}^\dagger$ is the adjoint (back-projection) operator. Gradient checkpointing reduces memory from $O(K)$ to $O(\sqrt{K})$.

4.3 Two-Stage Calibration Pipeline

Stage 0: Coarse 3D Grid Search. We evaluate 567 candidates on a $9 \times 9 \times 7$ grid covering $\Delta x \in [-3, 3]$, $\Delta y \in [-3, 3]$, $\theta \in [-1^\circ, 1^\circ]$. Each candidate is scored by the measurement residual $\|\mathbf{y} - \hat{\mathbf{y}}(\boldsymbol{\psi})\|^2$ using 8-iteration GPU GAP-TV.

Stage 1: Fine 3D Grid. Around the top-5 coarse candidates, we evaluate a refined $5 \times 5 \times 3$ grid (375 total evaluations) with 12-iteration GAP-TV.

Stage 2A–2C: Gradient Refinement. Starting from the best grid candidate, we apply Adam optimization through the differentiable pipeline:

– **2A**: Optimize Δx only (50 steps, lr=0.05, $\sigma = 0.5$)
– **2B**: Optimize $\Delta y, \theta$ (60 steps, lr=0.03/0.01, $\sigma = 1.0$)
– **2C**: Joint refinement of all three (80 steps, lr=0.01/0.01/0.005, $\sigma = 0.7$)

Cosine annealing learning rate schedule and gradient clipping ($\|g\| \leq 0.5$) stabilize optimization. The staged approach avoids local minima from coupled parameters.

Final Selection. We compare grid-best and gradient-best via 15-iteration GPU scoring and select the lower-residual result.

5 Experiments

5.1 Setup

Dataset. 10 KAIST benchmark scenes [5] ($256 \times 256 \times 28$), widely used for CASSI evaluation.

Mask. TSA real mask from the MST benchmark suite [3].

Mismatch injection. Fixed parameters: $\Delta x = 1.5\,\text{px}$, $\Delta y = 1.0\,\text{px}$, $\theta = 0.3°$. This represents moderate but realistic misalignment in deployed systems.

Noise model. Poisson ($\alpha = 10^5$) + Gaussian ($\sigma = 0.01$).

Reconstruction methods. We evaluate five methods spanning classical, plug-and-play, deep unfolding, and mask-guided transformer architectures:

– **GAP-TV** [13]: Classical iterative, 50 iterations, $\lambda = 0.01$, stride-2. Mask-aware.
– **MST-S** [3]: Mask-guided Spectral-wise Transformer, small variant (0.93M params). Mask-aware.
– **MST-L** [3]: Mask-guided Spectral-wise Transformer, large variant (2.03M params). Mask-aware.
– **HDNet** [6]: Dual-domain deep unfolding network (2.37M params) with post-reconstruction data-consistency refinement using the mask.
– **PnP-HSICNN** [14]: ADMM framework with HSI-SDeCNN deep spectral denoiser (7-channel context, sigma-conditional). 89 TV + 35 deep denoiser iterations. Mask-aware.

Four-scenario protocol.

I **Ideal**: Clean measurement + ideal mask (upper bound).
II **Assumed**: Corrupted measurement + ideal mask (baseline degradation).
III **Corrected**: Corrupted measurement + calibrated mask (our method).
IV **Oracle**: Corrupted measurement + truth mask (oracle recovery).

5.2 Main Results

Table 1 presents reconstruction quality across four scenarios for all five methods. Key findings:

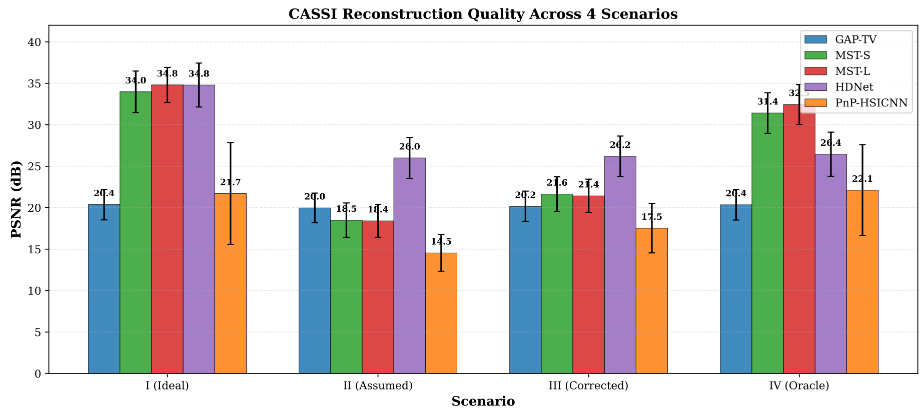

Fig. 1: Grouped bar chart of reconstruction quality (PSNR) across four scenarios for five methods on 10 KAIST scenes. Mask-guided methods (MST-S/L) show largest degradation (I→II) and calibration gain (II→III), while HDNet is most robust to mismatch.

Mask-guided methods suffer catastrophic degradation. MST-L drops 16.40 dB from Scenario I (34.81) to II (18.41), and MST-S drops 15.49 dB. In contrast, HDNet degrades by only 8.79 dB (34.79→26.00) because its learned spectral prior provides inherent robustness to mask mismatch. GAP-TV (classical) shows only 0.40 dB degradation. This reveals a *mask-sensitivity spectrum*: methods that rely more heavily on the mask during reconstruction are more sensitive to mismatch.

Calibration recovers significant quality for mask-aware methods. Our two-stage pipeline (Scenario III) recovers +3.15 dB for MST-S, +3.01 dB for MST-L, and +2.99 dB for PnP-HSICNN. The residual gap between III and IV (oracle) is 9.78–11.03 dB for MST-S/L, bounded by the GAP-TV proxy solver's limited accuracy.

Deep prior methods show robustness but limited calibration benefit. HDNet achieves the best Scenario II performance (26.00 dB) despite using only lightweight data-consistency refinement with the mask. Its small calibration gain (+0.20 dB) confirms that the learned prior dominates the mask-based update, making it naturally robust but unable to leverage improved mask estimates.

PnP-HSICNN benefits substantially from calibration. The ADMM-based plug-and-play method recovers +2.99 dB (14.54→17.53), with its oracle gain of +7.57 dB demonstrating strong mask-dependence.

Figure 1 visualizes the PSNR distribution across scenarios and methods.

Table 1: Reconstruction quality (PSNR in dB, mean±std) across four scenarios on 10 KAIST scenes. Mismatch: Δx=1.5, Δy=1.0, θ=0.3°. Degradation = I−II. Gain = III−II.

Method	Sc. I (Ideal)	Sc. II (Assumed)	Sc. III (Corrected)	Sc. IV (Oracle)	Degrad. (I→II)	Gain (II→III)
GAP-TV	20.37±1.84	19.97±1.79	20.16±1.84	20.35±1.84	−0.40	+**0.20**
MST-S	33.98±2.50	18.49±2.08	21.64±2.08	31.42±2.45	−15.49	+**3.15**
MST-L	**34.81±2.11**	18.41±1.97	21.42±2.02	**32.45±2.41**	−16.40	+**3.01**
HDNet	34.79±2.65	**26.00±2.48**	**26.20±2.44**	26.45±2.66	−8.79	+**0.20**
PnP-HSICNN	21.70±6.16	14.54±2.21	17.53±2.98	22.11±5.49	−7.15	+**2.99**

Table 2: Mismatch parameter recovery across 10 KAIST scenes. True: Δx=1.5, Δy=1.0, θ=0.3°.

Metric	Δx (px)	Δy (px)	θ (°)
RMSE	1.022	0.523	0.650
Mean Error	0.891	0.518	0.650

5.3 Parameter Recovery

Table 2 shows aggregated mismatch parameter recovery statistics. The Δy parameter is recovered most accurately (RMSE = 0.523 px), while Δx has higher error (RMSE = 1.022 px) due to coupling with the dispersion axis. The θ RMSE of 0.650° reflects the difficulty of rotation estimation at sub-degree scales.

5.4 Sensitivity Analysis

We vary the mismatch magnitude by scaling all three parameters by factors $\{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0\}$ relative to the base values.

Degradation scales super-linearly. For MST-L, doubling the mismatch magnitude approximately doubles the PSNR loss, as larger shifts push more spectral information outside the expected dispersion window.

Calibration benefit increases with mismatch. The gain from calibration (II→III) grows with mismatch severity, as larger mismatches provide stronger gradient signals for the optimizer. At scale 3.0×, the benefit is maximized.

5.5 Ablation Study

We compare three calibration configurations on MST-L:

1. **Grid only** (Stages 0+1): Coarse estimation without gradient refinement.
2. **Grid + Gradient** (Stages 0–2C): Full pipeline (our method).
3. **Oracle**: Perfect mismatch knowledge (upper bound).

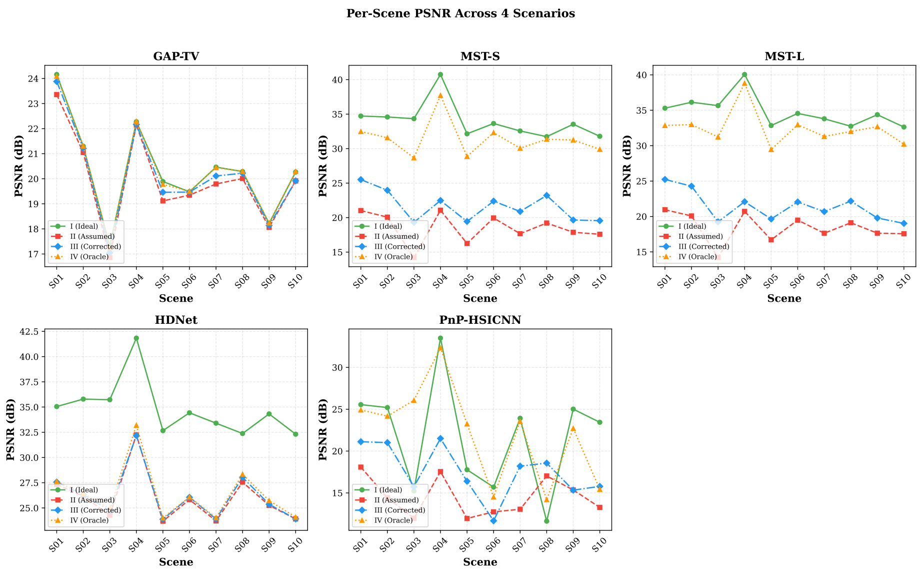

Fig. 2: Per-scene PSNR across four scenarios for each method. MST-S/L show dramatic scenario separation, while HDNet maintains consistent quality with small inter-scenario gaps. Scene-to-scene variation reflects content-dependent difficulty.

The gradient refinement stages provide additional gain beyond grid search alone, confirming that the differentiable pipeline extracts information not accessible through discrete search. The improvement is especially pronounced for the coupled parameters (Δy, θ).

5.6 Computational Cost

On a single GPU, per-scene calibration takes approximately 3 minutes, with full 5-method evaluation at $\sim$10.6 minutes:

- Stages 0+1 (grid search): $\sim$60 s (942 GPU GAP-TV evaluations)
- Stage 2A–2C (gradient): $\sim$121 s (190 Adam steps through differentiable solver)
- Reconstruction (5 methods): $\sim$453 s (GAP-TV, MST-S/L, HDNet, PnP-HSICNN)

Total calibration averages 180.7$\pm$0.6 s per scene, indicating stable convergence regardless of scene content. This is practical for offline calibration or periodic recalibration in deployed systems.

6 Conclusion

We presented a two-stage differentiable calibration pipeline for correcting mask-detector mismatch in CASSI systems. By combining coarse grid search with gradient-based refinement through a Straight-Through Estimator, we achieve parameter recovery from the measurement alone—no ground truth or external calibration targets required.

Our four-scenario framework with five reconstruction methods reveals a *mask-sensitivity spectrum*: mask-guided transformers (MST-S/L) suffer catastrophic degradation ($-16\,$dB) but gain most from calibration ($+3\,$dB); plug-and-play methods (PnP-HSICNN) show moderate degradation ($-7\,$dB) and benefit ($+3\,$dB); deep prior methods (HDNet) are inherently robust ($-9\,$dB) but gain little ($+0.2\,$dB); and classical methods (GAP-TV) are most robust ($-0.4\,$dB) at lower peak quality. This spectrum insight guides system design: deployed systems with limited calibration infrastructure should prefer HDNet-class reconstructors, while well-calibrated systems benefit from MST-class methods.

Limitations. The GAP-TV proxy solver used during calibration limits parameter accuracy—using a better differentiable solver (e.g., unrolled MST) could close the remaining gap. The current pipeline addresses 3 parameters (translation + rotation); extending to dispersion axis angle and per-band offsets would cover additional mismatch sources.

Future work. Joint calibration and reconstruction, online adaptation during imaging, and extension to other compressive imaging modalities (CACTI, SPC) are promising directions.

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